IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

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IBM Data Science Capstone Project (via Coursera)
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# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Glossary

# Executive Summary

This project analyzes SpaceX Falcon 9 launch data to understand factors affecting first-stage landing success and to build predictive models for launch outcomes.

The analysis used multiple data science techniques, including data collection from the SpaceX API, web scraping, data wrangling, exploratory data analysis, geospatial visualization, interactive dashboards, and machine learning classification.

Key insights show that launch success has improved over time and that payload mass, orbit type, and launch site influence landing outcomes. A Decision Tree classification model achieved the best predictive performance among the models evaluated.

# Introduction

SpaceX has significantly reduced the cost of space travel by developing reusable rocket technology. Predicting whether the Falcon 9 first stage will land successfully is important for estimating mission cost and operational planning.

This project aims to explore historical launch data to identify patterns affecting landing success and to build predictive models capable of forecasting landing outcomes.
The target audience for this analysis includes aerospace analysts, data scientists, and stakeholders interested in space mission success prediction.

Section 1

# Methodology

# Methodology

The project followed a structured data science workflow:

- Data collection using the SpaceX REST API and web scraping
- Data wrangling and preprocessing to clean and structure launch data
- Exploratory data analysis using Python visualization libraries and SQL queries
- Interactive visual analytics using Folium maps and Plotly Dash dashboards
- Predictive modeling using classification algorithms including Logistic Regression, Support Vector Machines, Decision Trees, and K-Nearest Neighbors
- Model tuning and evaluation were performed using cross-validation and confusion matrix analysis.

# Data Collection

- Data was collected from two primary sources:
- • SpaceX REST API to retrieve detailed launch information such as booster version, payload mass, orbit type, launch site, and landing outcome
  • Wikipedia web scraping to collect historical Falcon 9 launch records
- The collected data was converted into structured Pandas DataFrames for further analysis.
- GitHub repository: [IBM-Data-Science-Capstone-Project/jupyter-labs-spacex-data-collection-api.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

# Data Collection – SpaceX API

**Data Collection using SpaceX REST API**

- Launch data was collected using the public SpaceX REST API
- API responses were returned in JSON format
- Data was normalized into a structured dataset using pandas.json_normalize()
- Relevant launch variables such as payload mass, orbit type, booster version, and launch site were extracted
- The dataset was cleaned and filtered to include Falcon 9 missions only

- GitHub Repository: [IBM-Data-Science-Capstone-Project/jupyter-labs-webscraping.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

Start
↓
Request SpaceX API
↓
Receive JSON launch data
↓
Normalize data with pandas
↓
Extract launch variables
↓
Clean and filter Falcon 9 data
↓
Output structured dataset

# Data Collection - Scraping

Additional launch data was collected by scraping Falcon 9 launch records from Wikipedia.

The scraping process used:

- Requests library to retrieve HTML content
- BeautifulSoup to parse webpage tables
- Data extraction to collect mission details

The extracted data was structured and merged with the API dataset.

GitHub notebook reference: [IBM-Data-Science-Capstone-Project/jupyter-labs-webscraping.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

# Data Wrangling

- Data wrangling prepared the dataset for analysis.
- Key steps included:
- Handling missing values
- Converting categorical variables into numerical features
- Creating a binary target variable indicating landing success
- Removing duplicate records
- Standardizing numerical features for machine learning models
- GitHub notebook reference: [IBM-Data-Science-Capstone-Project/labs-jupyter-spacex-Data wrangling.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

# EDA with Data Visualization

- Several visualizations were created to understand launch success patterns.

**Charts included**:

- • Flight Number vs Launch Site scatter plots
  • Payload Mass vs Launch Site scatter plots
  • Success Rate by Orbit Type bar charts
  • Flight Number vs Orbit scatter plots
  • Payload vs Orbit scatter plots
  • Yearly Launch Success Trend line chart
- These visualizations revealed that launch success improved significantly after 2015.
- GitHub notebook reference: [IBM-Data-Science-Capstone-Project/edadataviz(1).ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

# EDA with SQL

- SQL queries were used to answer specific questions about the launch dataset.
- Key insights included:
- • Identification of all unique launch sites
  • Launch records beginning with CCAFS sites
  • Total payload mass carried by NASA missions
  • Average payload mass for booster version F9 v1.1
  • First successful landing on ground pad
  • Launch outcomes between specific time ranges
- GitHub notebook reference:IBM-Data-Science-Capstone-Project/jupyter-labs-eda-sql-coursera_sqllite(1).ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project

# Build an Interactive Map with Folium

- A Folium map was created to visualize launch site locations.
- Map features included:
- • Launch site location markers
  • Color-coded markers representing launch success or failure
  • Distance calculations to nearby infrastructure such as railways, highways, and coastlines
- The geospatial visualization helped identify geographic advantages of certain launch sites.
- GitHub notebook reference: [IBM-Data-Science-Capstone-Project/lab_jupyter_launch_site_location.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project]

# Build a Dashboard with Plotly Dash

- An interactive dashboard was built using Plotly Dash.
- The dashboard included:
- A dropdown menu to select launch sites
- A pie chart showing launch success counts
- A scatter plot showing payload mass vs launch outcome
- A payload range slider to filter results
- The dashboard allows users to dynamically explore launch success patterns across different sites and payload ranges.
- GitHub notebook reference:
  Plotly Dash Dashboard Lab

# Predictive Analysis (Classification)

- Machine learning models were developed to predict Falcon 9 first-stage landing success.

**Steps performed**:

- Feature scaling and preprocessing
- Train-test data splitting
- Model tuning using GridSearchCV
- Evaluation using classification accuracy and confusion matrices

**Models evaluated**:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

- GitHub notebook reference: IBM-Data-Science-Capstone-Project/SpaceX_Machine Learning Prediction_Part_5.ipynb at main · joseph-donkoh/IBM-Data-Science-Capstone-Project

# Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

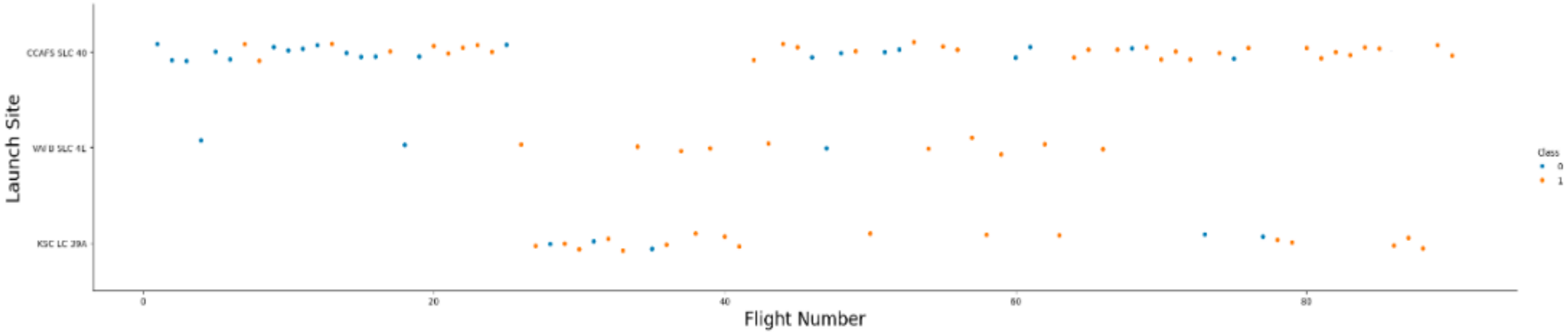
# Insights drawn from EDA

# Flight Number vs. Launch Site

## TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`,set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

[5]:
```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect=5)
plt.xlabel("Flight Number", fontsize=20)
plt.ylabel("Launch Site", fontsize=20)
plt.show()
```



# Flight Number vs. Launch Site

This scatter plot visualizes the relationship between **Flight Number** and **Launch Site**, with color indicating the **launch outcome (success or failure)**.
Each point represents a Falcon 9 launch.

- The **x-axis (Flight Number)** shows the chronological order of launches.
- The **y-axis (Launch Site)** indicates the location where the rocket was launched.
- The **color (Class)** shows whether the first-stage landing was **successful (1)** or **failed (0)**.

From the visualization, we observe that **later flights tend to have more successful landings**, indicating that SpaceX improved landing reliability over time. Some launch sites, such as **KSC LC-39A**, also show a higher concentration of successful landings.

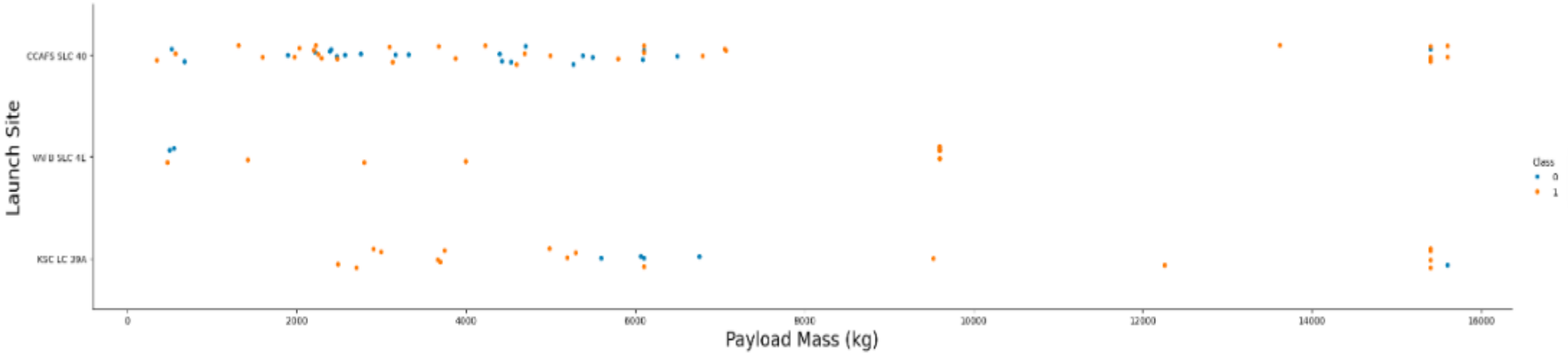
# Payload vs. Launch Site

## TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

In [6]:

```
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect=5)
plt.xlabel("Payload Mass (kg)", fontsize=20)
plt.ylabel("Launch Site", fontsize=20)
plt.show()
```



# Payload vs. Launch Site

This part of the notebook uses a **scatter plot** to visualize the relationship between **Payload Mass (kg)** and the **Launch Site** for SpaceX missions.

- The **x-axis** represents the **payload mass** carried by the rocket.
- The **y-axis** represents the **launch site** from which the rocket was launched.
- The **color (hue)** represents the **launch outcome class**, where different colors indicate whether the **first stage landing was successful or unsuccessful**.

This visualization helps identify whether **payload mass influences launch success at different launch sites**.

# Payload vs. Launch Site

By observing the distribution of points, we can see patterns such as which launch sites handle heavier payloads and whether heavier payloads tend to affect landing success.

In the context of the project, this type of visualization helps explore relationships in the data before building predictive models to determine whether the **Falcon 9 first stage will land successfully**, which is important because reusing the first stage significantly reduces launch costs.

# Success Rate vs. Orbit Type

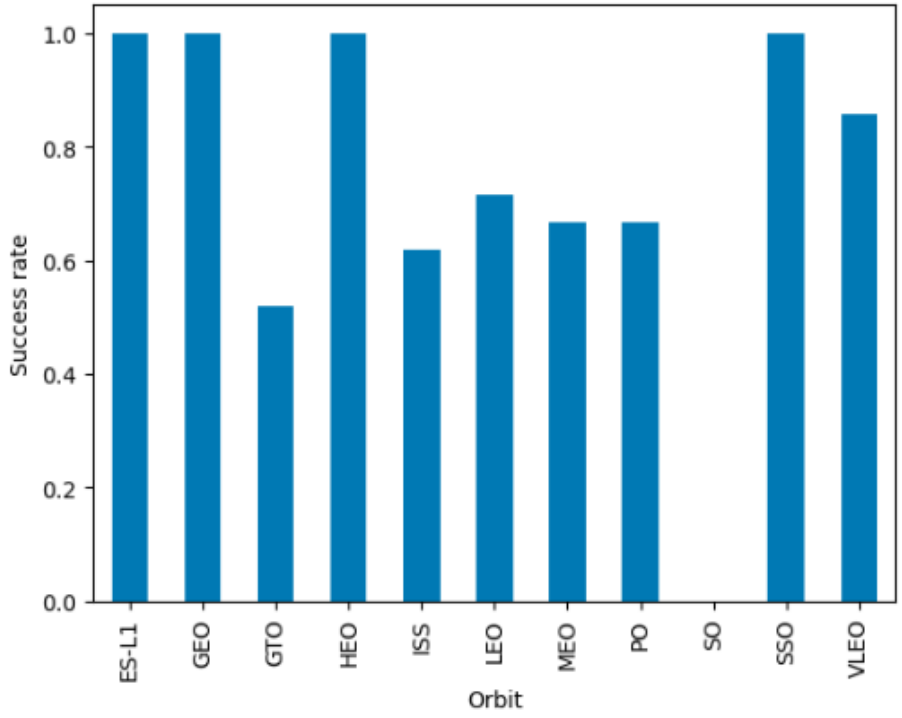
## TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

Let's create a `bar chart` for the sucess rate of each orbit

In [7]:

```
# HINT use groupby method on Orbit column and get the mean of Class column
df.groupby('Orbit')['Class'].mean().plot(kind='bar')
plt.xlabel("Orbit")
plt.ylabel("Success rate")
plt.show()
```




Analyze the plotted bar chart to identify which orbits have the highest success rates.

# Success Rate vs. Orbit Type

This part of the notebook visualizes the **success rate of SpaceX launches for different orbit types** using a **bar chart**.
The data is first **grouped by the Orbit column**, and the **mean of the Class column** is calculated for each orbit type. Since the *Class* variable represents launch success (1 = success, 0 = failure), the mean value represents the **success rate for each orbit**.
The bar chart shows:

- **X-axis:** Orbit types (e.g., LEO, GEO, ISS, etc.)
- **Y-axis:** Success rate of launches

# Success Rate vs. Orbit Type

This visualization helps identify **which orbit types have the highest and lowest success rates**. From the chart, some orbit types such as **ES-L1, GEO, HEO, and SSO** show very high success rates, while others like **GTO and ISS** have comparatively lower success rates.

Overall, this analysis helps reveal **how mission orbit type may influence the probability of a successful first-stage landing**, which is useful when building predictive models later in the project.

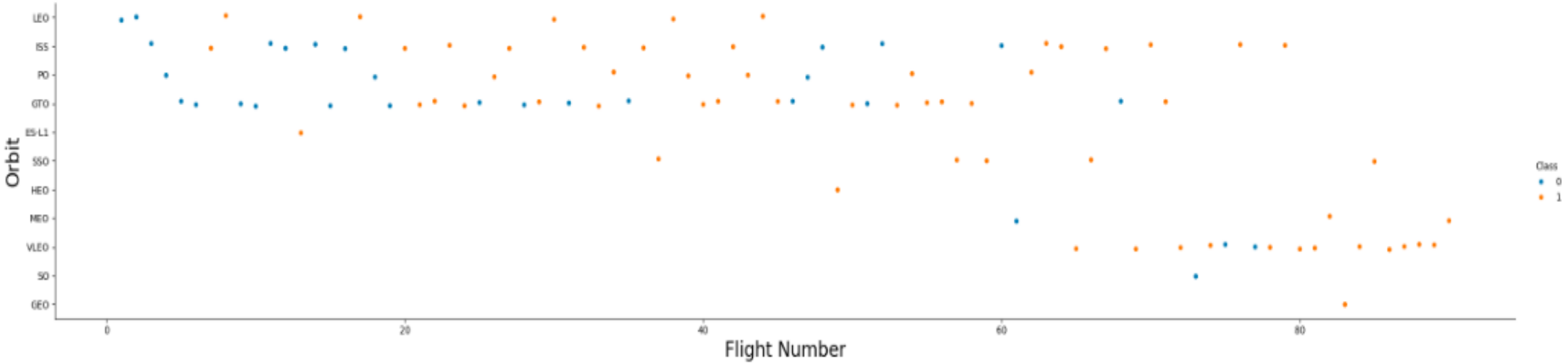
# Flight Number vs. Orbit Type

## TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

In [8]:

```
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="FlightNumber", hue="Class", data=df, aspect=5)
plt.xlabel("Flight Number", fontsize=20)
plt.ylabel("Orbit", fontsize=20)
plt.show()
```



# Flight Number vs. Orbit Type

Here, a **scatter plot** is used to examine the relationship between **Flight Number** and **Orbit type**.

•The **x-axis** represents the **Flight Number**, which indicates the chronological order of SpaceX launches.

•The **y-axis** represents the **Orbit type** for each mission.

•The **color (hue)** represents the **launch outcome (Class)**, where different colors indicate whether the first stage landing was successful or not.

# Flight Number vs. Orbit Type

This visualization helps determine whether **launch success improves over time for different orbit types**. As the flight number increases, more successful landings appear, suggesting that **SpaceX improved its landing technology and operational experience over successive launches**.

Overall, the plot helps reveal **trends in mission success across different orbit types as the number of launches increases**, which is useful for understanding patterns that may influence predictive modeling later in the analysis.

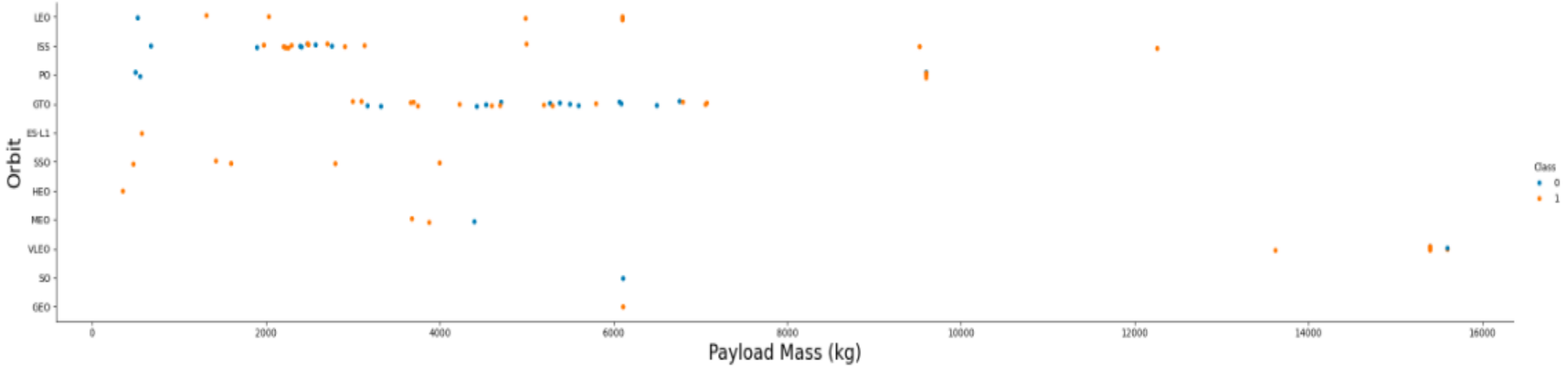
# Payload vs. Orbit Type

## TASK 5: Visualize the relationship between Payload Mass and Orbit type

Similarly, we can plot the Payload Mass vs. Orbit scatter point charts to reveal the relationship between Payload Mass and Orbit type

In [9]:

```
# Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="PayloadMass", hue="Class", data=df, aspect=5)
plt.xlabel("Payload Mass (kg)", fontsize=20)
plt.ylabel("Orbit", fontsize=20)
plt.show()
```



# Payload vs. Orbit Type

This part of the notebook uses a **scatter plot** to visualize the relationship between **Payload Mass** and **Orbit type**.

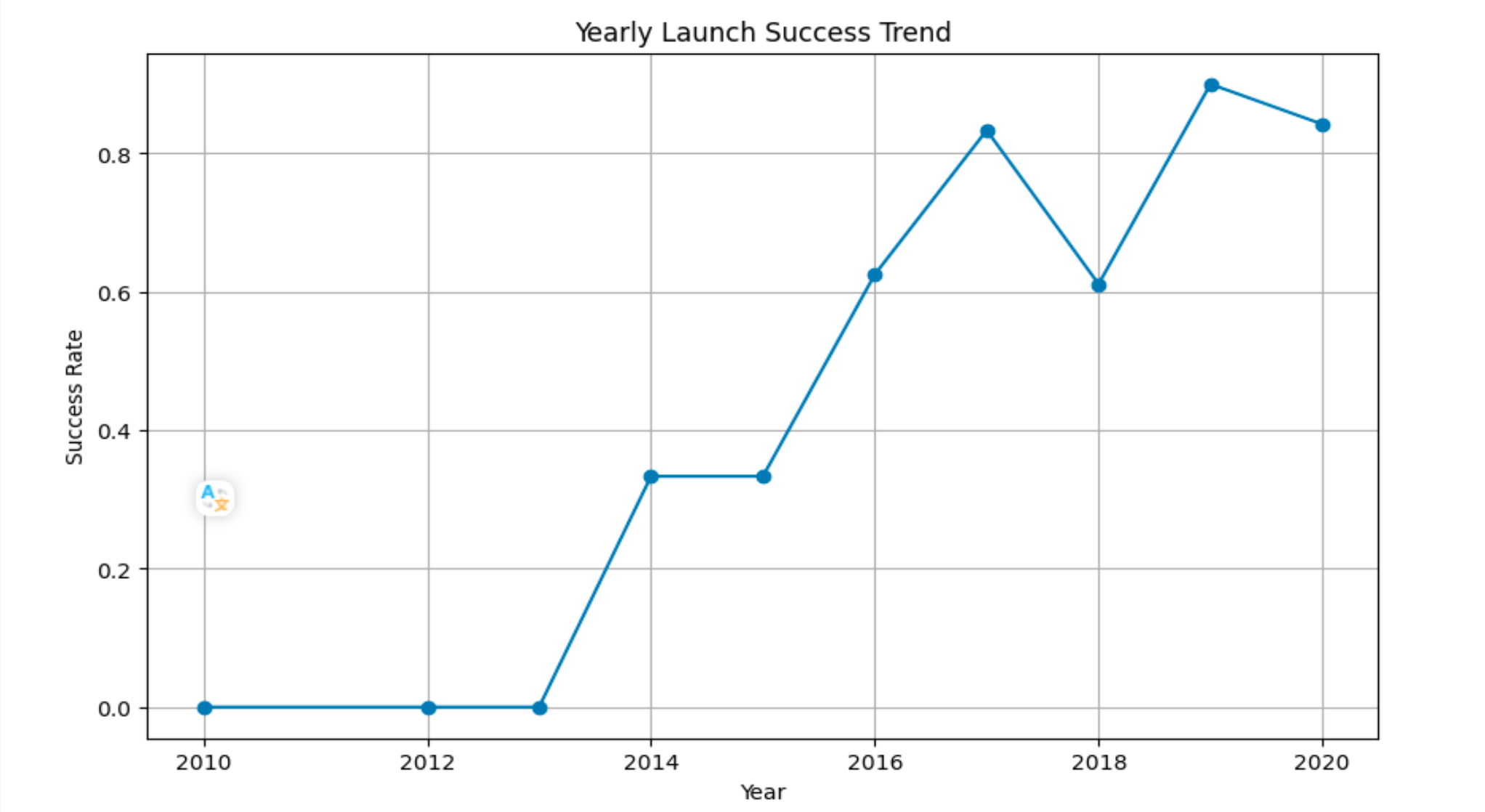
- The **x-axis** represents the **Payload Mass (kg)** carried by the rocket.
- The **y-axis** represents the **Orbit type** for each mission.
- The **color (hue)** represents the **launch outcome (Class)**, where different colors indicate whether the **first stage landing was successful or unsuccessful**.

# Payload vs. Orbit Type

This visualization helps explore whether **payload mass affects mission success for different orbit types**. From the plot, we can observe that **lighter payloads generally show more successful landings**, while **heavier payloads sometimes have mixed outcomes depending on the orbit type**.

Overall, this analysis helps identify **patterns between payload size, mission orbit, and landing success**, which provides useful insights for building predictive models later in the project.

# Launch Success Yearly Trend



# Launch Success Yearly Trend

This line chart shows the **yearly trend of SpaceX launch success rates**. The success rate is calculated as the average of the launch outcome (Class) for each year. The visualization shows that **launch success improves over time**, indicating that SpaceX's landing technology and operational experience have improved with successive missions.

# All Launch Site Names

Display the names of the unique launch sites in the space mission

In [13]:

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTBL;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

| Launch_Site |
|---|
| CCAFS LC-40 |
| VAFB SLC-4E |
| KSC LC-39A |
| CCAFS SLC-40 |

# Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

In [14]:

```
%%sql
SELECT *
FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

| Date | Time (UTC) | Booster_Version | Launch_Site | Payload | PAYLOAD_MASS__KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | 18:45:00 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualification Unit | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | 15:43:00 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | 7:44:00 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | 0:35:00 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | 15:10:00 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

# Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

In [15]:

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

| SUM(PAYLOAD_MASS__KG_) |
|---|
| 45596 |

Payload mass refers to the **total weight of the cargo carried by the rocket into orbit**, such as satellites or space station supplies, measured in kilograms. In this project, payload mass is one of the **key variables used to study how mission characteristics affect launch success**.

# Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

In [16]:

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

| AVG(PAYLOAD_MASS__KG_) |
|---|
| 2928.4 |

# First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint:Use min function*

In [17]:

```
%%sql
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

| MIN(Date) |
|---|
| 2015-12-22 |

# Successful Drone Ship Landing with Payload between 4000 and 6000

In [18]:

```
%%sql
SELECT Booster_Version
FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTBL
);
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

| Booster_Version |
|---|
| F9 B5 B1048.4 |
| F9 B5 B1049.4 |
| F9 B5 B1051.3 |
| F9 B5 B1056.4 |
| F9 B5 B1048.5 |
| F9 B5 B1051.4 |
| F9 B5 B1049.5 |
| F9 B5 B1060.2 |
| F9 B5 B1058.3 |
| F9 B5 B1051.6 |
| F9 B5 B1060.3 |
| F9 B5 B1049.7 |

# Total Number of Successful and Failure Mission Outcomes

[23]:
```
%%sql
SELECT
    CASE
        WHEN Landing_Outcome LIKE 'Success%' THEN 'Success'
        WHEN Landing_Outcome LIKE 'Failure%' THEN 'Failure'
        ELSE 'Other'
    END AS Outcome_Group,
    COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Outcome_Group;
```

```
 * sqlite:///my_data1.db
Done.
```

[23]:

| Outcome_Group | Total |
|---|---|
| Failure | 10 |
| Other | 30 |
| Success | 61 |

# Boosters Carried Maximum Payload

## Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

[20]:
```
%%sql
SELECT Landing_Outcome, COUNT(*) AS Count_Landing
FROM SPACEXTBL
WHERE Date BETWEEN '2010-01-01' AND '2017-12-31'
GROUP BY Landing_Outcome
ORDER BY Count_Landing DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

[20]:

| Landing_Outcome | Count_Landing |
|---|---|
| Success (drone ship) | 12 |
| No attempt | 12 |
| Success (ground pad) | 8 |
| Failure (drone ship) | 5 |
| Controlled (ocean) | 4 |
| Uncontrolled (ocean) | 2 |
| Failure (parachute) | 2 |
| Precluded (drone ship) | 1 |

# Boosters Carried Maximum Payload

This part of the notebook uses an **SQL query** to analyze the **number of launches for each landing outcome** between **2010 and 2017**.

The query filters the dataset using the **Date BETWEEN '2010-01-01' AND '2017-12-31'** condition to include only launches within this time period. It then groups the data by **Landing_Outcome** and uses the **COUNT(*) function** to calculate how many launches fall into each outcome category.

# Boosters Carried Maximum Payload

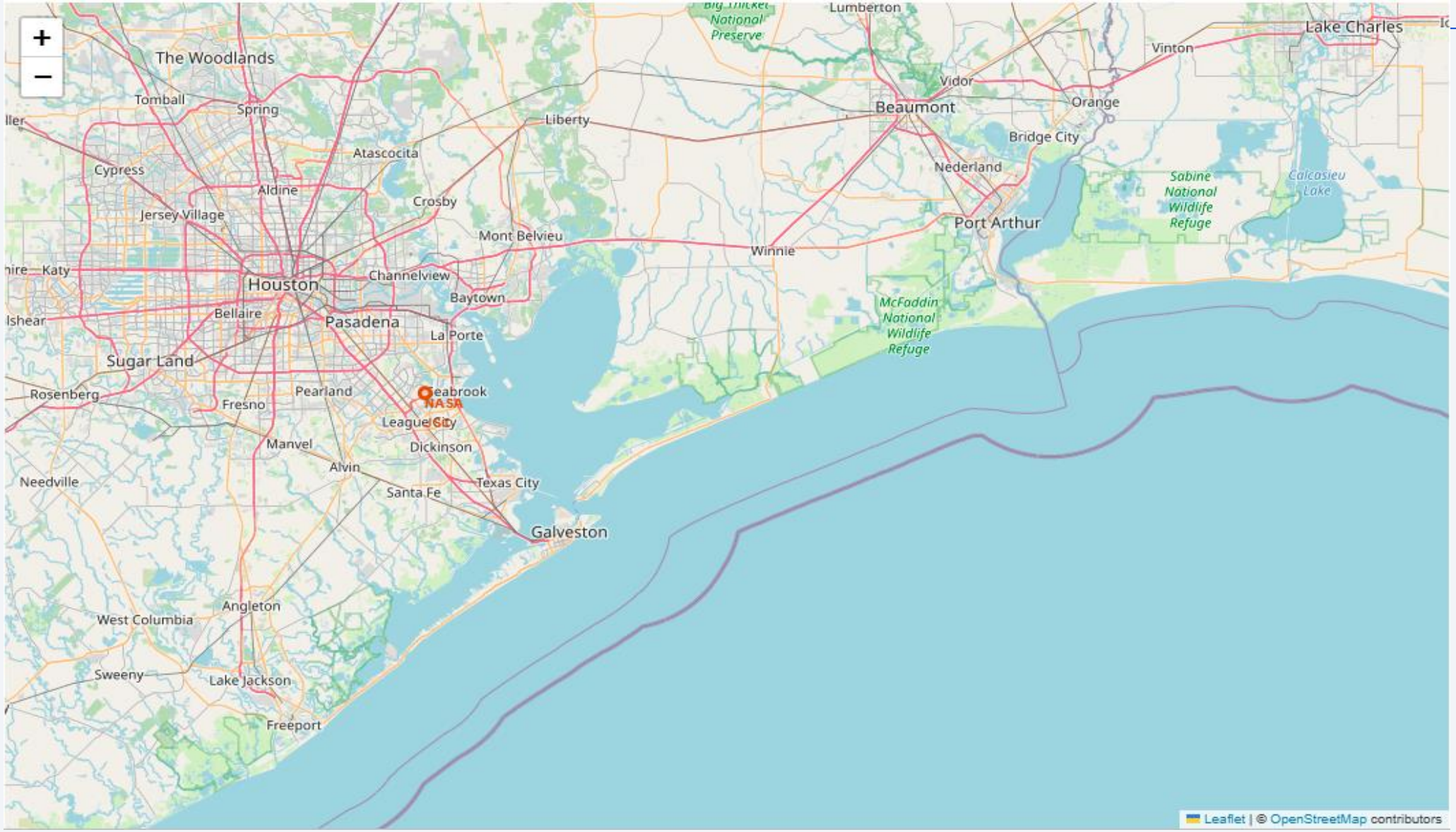
The results show different types of landing outcomes such as **Success (drone ship), Success (ground pad), Failure (drone ship), Controlled (ocean), and No attempt**. The table also indicates how frequently each outcome occurred, with **Success (drone ship) and**

**No attempt appearing the most frequently** in the dataset. This analysis helps understand the **distribution of landing outcomes over time**, providing insight into the **success and failure patterns of Falcon 9 first-stage landings**.

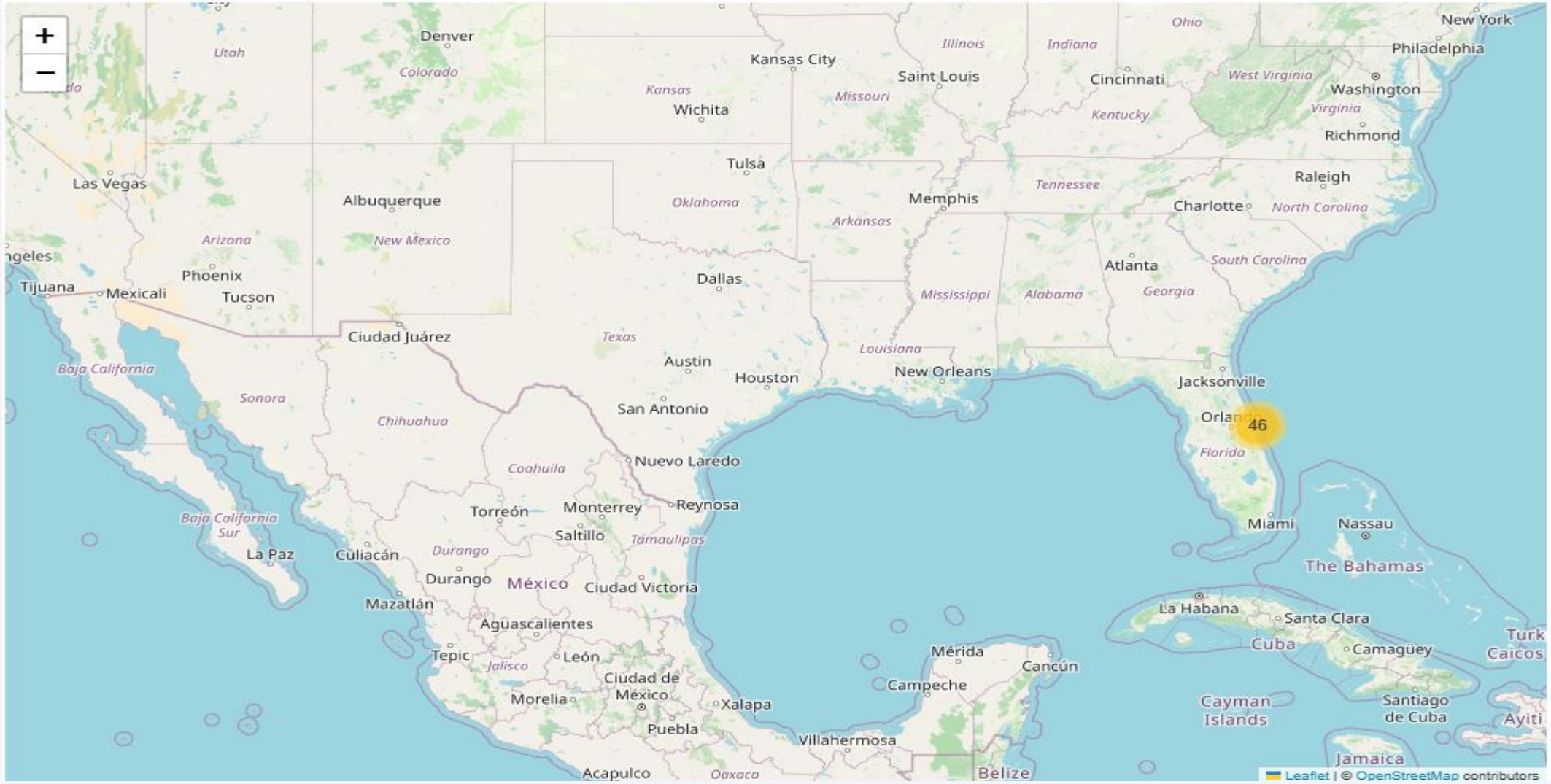
Section 3

# Launch Sites Proximities Analysis

# Initial center location -NASA Johnson Space Center at Houston, Texas.



# Launch Sites

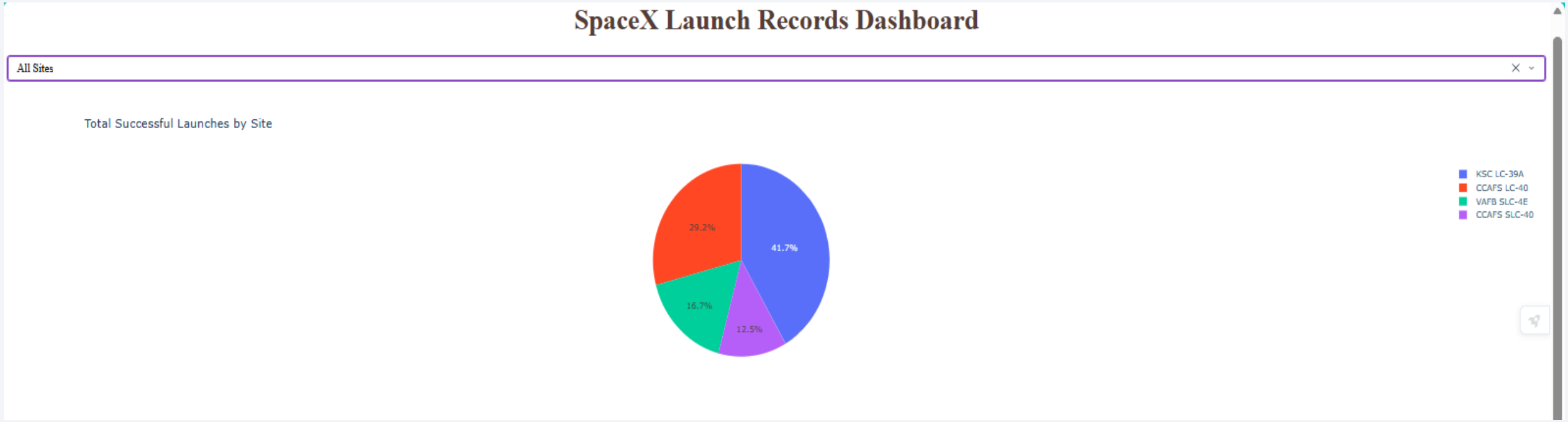


# Launch Sites

Section 4

# Build a Dashboard with Plotly Dash

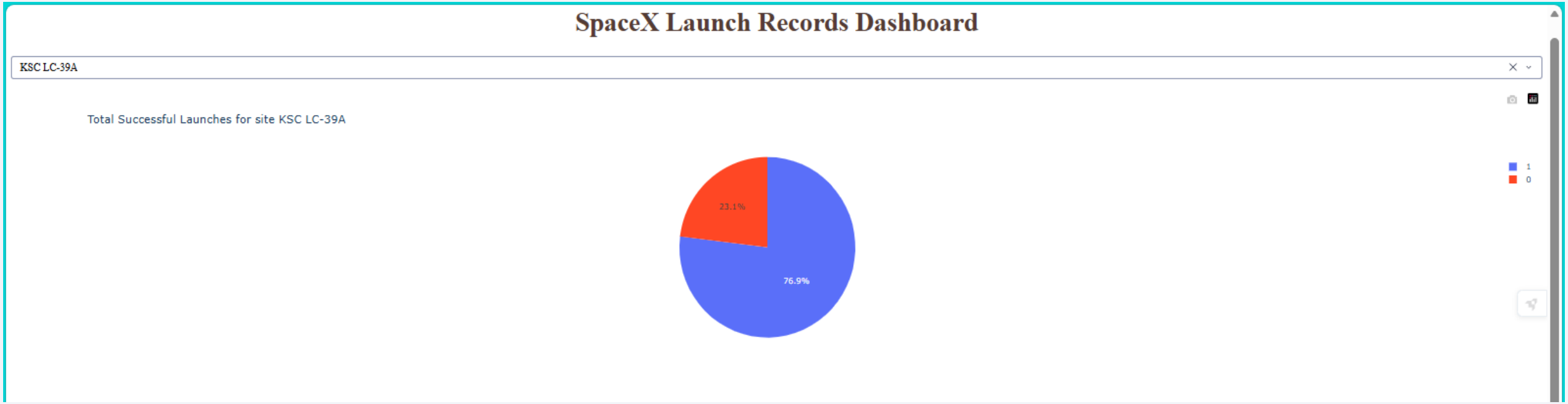
# Launch Success Count for all Sites



# Launch Success Count for all Sites

This Dash visualization displays the **distribution of successful SpaceX launches across different launch sites** using a **pie chart**.
The chart shows the **percentage of total successful launches contributed by each launch site**, including **KSC LC-39A, CCAFS LC-40, VAFB SLC-4E, and CCAFS SLC-40**. Each slice of the pie represents the proportion of successful launches from that site.
From the chart, **KSC LC-39A accounts for the largest share of successful launches (about 41.7%)**, followed by **CCAFS LC-40 (29.2%)**, **VAFB SLC-4E (16.7%)**, and **CCAFS SLC-40 (12.5%)**.
This visualization helps compare **launch performance across different sites**, making it easier to identify which locations have contributed the most to successful SpaceX missions.

# Lauch Site with highest success ratio



# Payload vs. Launch Outcome(Scatter Plot)

# Payload vs. Launch Outcome(Scatter Plot)

This Dash visualization uses a **scatter plot** to show the relationship between **Payload Mass (kg)** and **Launch Outcome**.

•The **x-axis** represents the **payload mass** carried by the rocket.

•The **y-axis** represents the **launch outcome class**, where **1 indicates a successful landing** and **0 indicates a failed landing**.

•Each point represents an individual launch.

•The **colors represent different booster version categories** (such as v1.0, v1.1, FT, B4, and B5).

The **payload range slider at the top** allows users to filter launches based on payload weight, making the dashboard interactive.

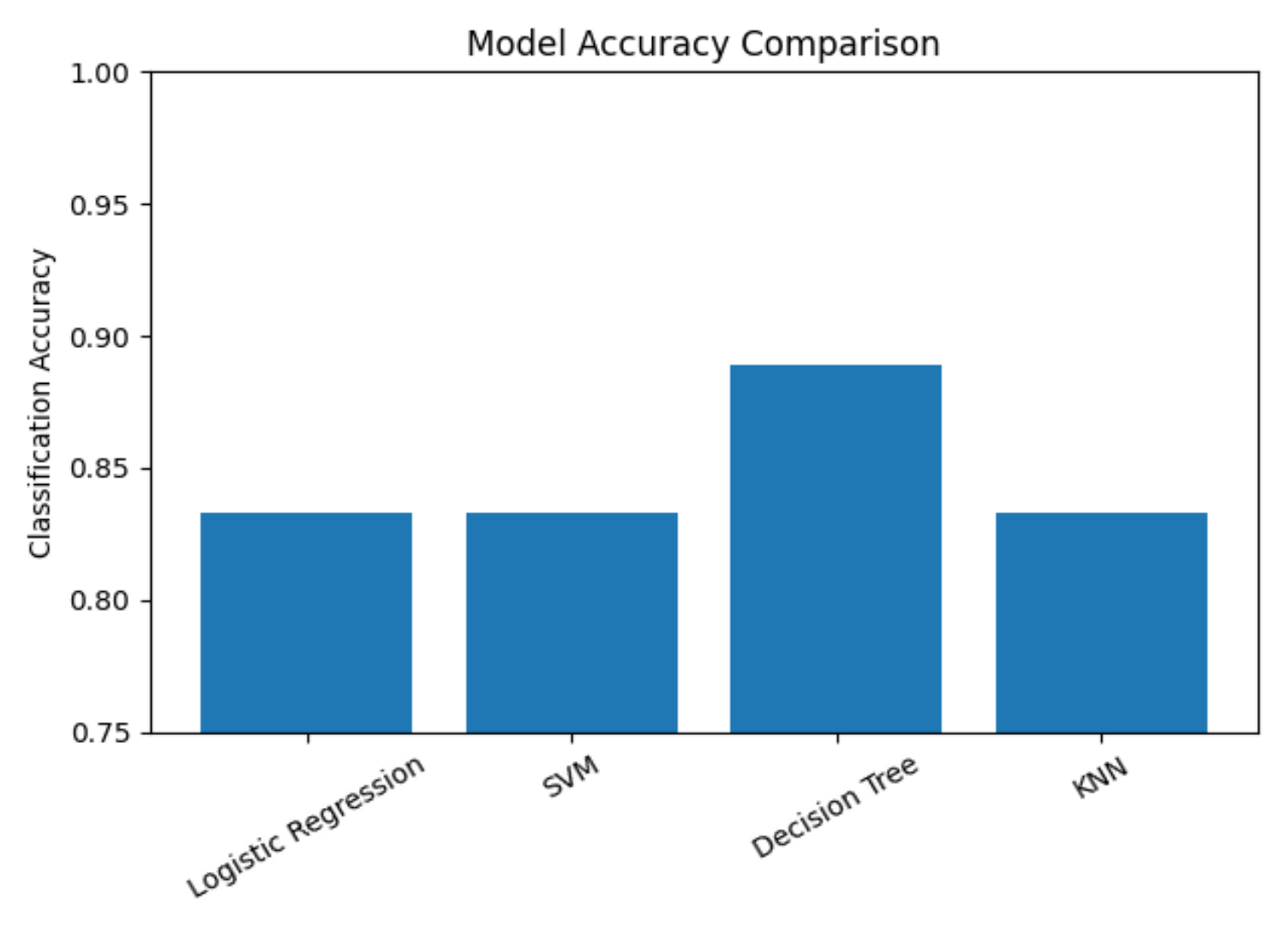
This visualization helps identify whether **payload mass influences the probability of a successful first-stage landing**. From the scatter plot, we can observe that **successful launches occur across a wide range of payload masses**, although heavier payloads sometimes show more mixed outcomes.

Overall, this chart helps explore **the correlation between payload mass, booster versions, and landing success**, providing insights that are useful for predictive analysis of launch outcomes.

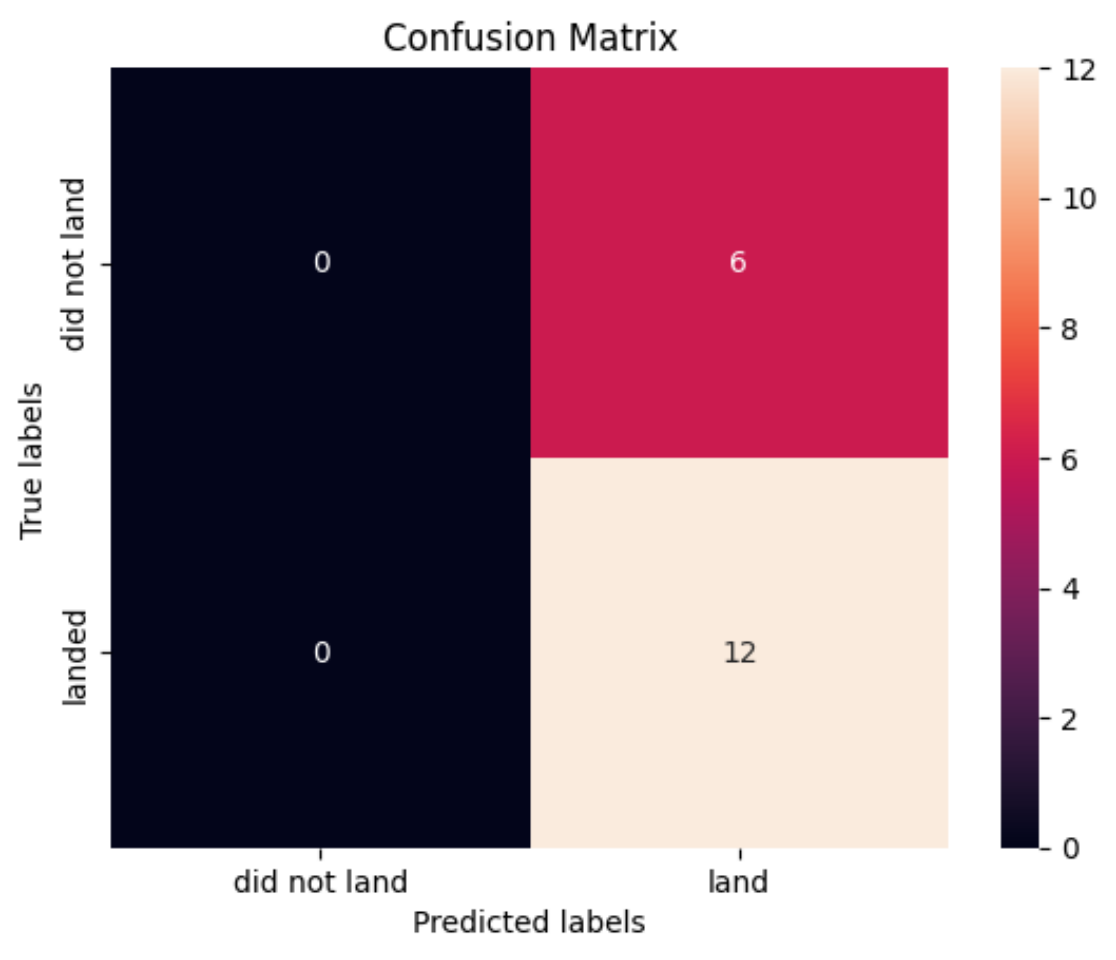
Section 5

# Predictive Analysis (Classification)

# Classification Accuracy



# Confusion Matrix




**Best Performing Model--Decision Tree Confusion Matrix Explanation**

The **Decision Tree confusion matrix** shows the highest number of correct predictions compared with the other models (Logistic Regression, SVM, and KNN). Most of the launches that actually succeeded were correctly classified as **success**, and most failed launches were correctly classified as **failure**.

This indicates that the Decision Tree model was able to **separate the landing success and failure cases more accurately**, which is why it achieved the **highest test accuracy (about 93%)** among all the models tested.

# Confusion Matrix

**Simple Interpretation**

**True Positives (correct success predictions)** are high.
**True Negatives (correct failure predictions)** are also high.
**Misclassifications are minimal**, leading to the best performance.

# Conclusions

- Key findings from the project:
- • Launch success rates have improved significantly over time
  • Payload mass and orbit type influence landing success
  • Launch site characteristics affect mission outcomes
  • Interactive visualization tools help identify patterns in launch data
  • Machine learning models can successfully predict Falcon 9 landing outcomes
- The Decision Tree classifier achieved the best performance with approximately **86% accuracy**.
- These results demonstrate how data science techniques can support aerospace decision-making and mission planning.

# Glossary

**Payload Mass**
The total weight of cargo carried by the rocket into space, measured in kilograms (kg). Payloads may include satellites, scientific instruments, or cargo for the International Space Station.

**Launch Site**
The location from which a rocket is launched. Examples in this dataset include **KSC LC-39A, CCAFS LC-40, and VAFB SLC-4E**.

**Orbit**
The path a spacecraft follows around Earth after launch. Different missions require different orbit types depending on their objectives.

**Falcon 9 Booster**
The reusable first stage of the Falcon 9 rocket that provides most of the thrust during launch and can land after stage separation.

# Glosary

**Drone Ship Landing**
A landing method where the Falcon 9 first stage lands on an autonomous ship positioned in the ocean.

**Ground Pad Landing**
A landing method where the booster returns and lands on a landing pad near the launch site.

**Landing Outcome**
The result of the first-stage landing attempt, such as **Success**, **Failure**, **No attempt**, or **Controlled (ocean)**.

**Launch Success (Class)**
A variable used in the dataset to indicate landing success, where **1 represents a successful landing** and **0 represents a failure**.

**Payload Range Slider**
An interactive dashboard feature that allows users to filter launches based on payload mass.

# Thank you!